

APAD PLATFORM — FINAL MASTER POC DOCUMENT

1. PROJECT TITLE

Authenticated Personalized Ad Delivery (APAD) Platform

Proof-of-Concept (POC) Demonstration System

2. PROJECT OBJECTIVE

The APAD platform is a proof-of-concept advertisement-driven authentication and personalized ad delivery platform.

The core idea of APAD is:

- Users must watch advertisements before receiving OTP authentication.
- Personalized advertisements are delivered using tokenized URLs and audience targeting.
- Authentication becomes an advertisement-gated workflow.
- Ads are displayed during the authentication process.
- Users are automatically logged into the portal after OTP verification.
- Personalized advertisements and offers are shown inside the portal.

This project is intended ONLY for:

- Investor demonstrations
- Patent validation
- Research prototype demonstrations
- UI/UX concept showcase
- Technical feasibility validation

This is NOT a production-grade platform.

3. CORE CONCEPT

Traditional systems:

User requests OTP
→ OTP sent immediately
→ User logs in

APAD system:

User requests OTP
→ User must watch advertisement
→ OTP generated only after ad completion
→ User enters OTP
→ User automatically logged in
→ Personalized ads shown inside portal

This creates:

- Advertisement-gated authentication
- Monetized OTP workflows
- Personalized ad delivery
- Authenticated advertising experiences

4. MAIN FEATURES

1. User Registration System

Users can:

- Self-register OR
- Be added by admin

User data includes:

- Name
 - Mobile Number
 - Age
 - Gender
 - Area/Location
-

2. Campaign Management System

Admin can:

- Create advertisement campaigns
- Upload creatives
- Configure targeting rules
- Manage campaigns

Campaign targeting includes:

- Age
 - Gender
 - Area
 - Interests (optional)
-

3. Audience Matching Engine

The backend automatically selects users matching campaign rules.

Example:

Campaign:

```
Travel Campaign
Target:
- Age 20-35
- Male
- Hyderabad
```

Matching Users:

```
John
Ravi
```

4. Tokenized Personalized Links

The system generates personalized secure links.

Example:

```
https://apad-demo.onrender.com/p/tk_92831
```

Each token maps to:

- User
- Campaign
- Advertisement
- Analytics

5. Dynamic Personalized Ad Previews

When messaging platforms fetch previews:

The backend dynamically generates:

```
<meta property="og:title"
content="John, Get 30% OFF on Bali">

<meta property="og:image"
content="travel_offer.jpg">

<meta property="og:description"
content="Exclusive offer for John">
```

This creates personalized previews for each user.

6. Advertisement-Gated OTP Authentication

Users MUST watch ads before OTP generation.

Workflow:

```
Watch Ad
→ Ad Completion Validation
→ OTP Generation
→ OTP Verification
→ Login Success
```

7. Multi-Surface Advertisement System

Ads are displayed across multiple authentication surfaces.

Examples:

Surface 1

Before OTP generation.

Surface 2

OTP confirmation screen.

Surface 3

OTP SMS promotional message.

Surface 4 (Optional)

OTP entry/login screen.

8. Auto Login System

After OTP verification:

- JWT/session created
 - User automatically logged in
 - Personalized portal loaded
-

9. Personalized Portal Experience

Inside portal:

- Personalized ads
 - Offers
 - Recommendations
 - Campaign engagement
 - User analytics
-

10. Analytics Tracking

Track:

- Ad impressions
 - Ad watch completion
 - Preview fetches
 - OTP requests
 - OTP verification
 - Login success
 - Conversions
 - Portal engagement
-

5. FLOW 1 — Personalized Ad Link → OTP → Portal Access

Flow Objective

Users receive personalized advertisement links on mobile.

Users must watch the advertisement completely before OTP is generated.

FLOW 1 MERMAID DIAGRAM

```
flowchart TD
```

```
A[Admin Creates Campaign]
```

```
--> B[Audience Matching Engine]
```

```
B --> C[Generate Tokenized Personalized Links]
```

```
C --> D[Send Personalized Ad Link to User Mobile]
```

```
D --> E[User Clicks Advertisement Link]
```

```
E --> F[Ad Surface 1<br/>Watch Advertisement Completely]
```

```
F --> G[Ad Completion Validation]
```

```
G --> H[Generate OTP]
```

```
H --> I[Ad Surface 2<br/>OTP Confirmation Screen + Ad]
I --> J[Send OTP SMS with Promotional Message]
J --> K[User Enters OTP]
K --> L[OTP Verification Success]
L --> M[Auto Login into Portal]
M --> N[Personalized Ads and Offers in Portal]
```

FLOW 1 SUMMARY

Main Concept

```
Ad First
→ OTP Later
→ Portal Access
```

Purpose

- Personalized campaign engagement
- Advertisement-based authentication
- Tokenized secure access
- User engagement tracking

6. FLOW 2 — Login OTP Flow with Ads Before OTP

Flow Objective

Users normally log into the application using mobile OTP.

However, users must watch advertisements BEFORE OTP generation.

FLOW 2 MERMAID DIAGRAM

```
flowchart TD
    A[User Opens APAD Application] --> B[Enter Mobile Number]
    B --> C[Click Get OTP]
    C --> D["Ad Surface 1<br/>Watch Ad Before OTP"]
    D --> E[Ad Completion Tracker]
    E --> F[Generate OTP]
    F --> G["Ad Surface 2<br/>OTP Sent Confirmation + Ad"]
    G --> H[Send OTP SMS with Promotional Ad]
    H --> I[User Enters OTP]
    I --> J[OTP Verification]
    J --> K[Login Success]
    K --> L[Personalized Portal Dashboard]
```

FLOW 2 SUMMARY

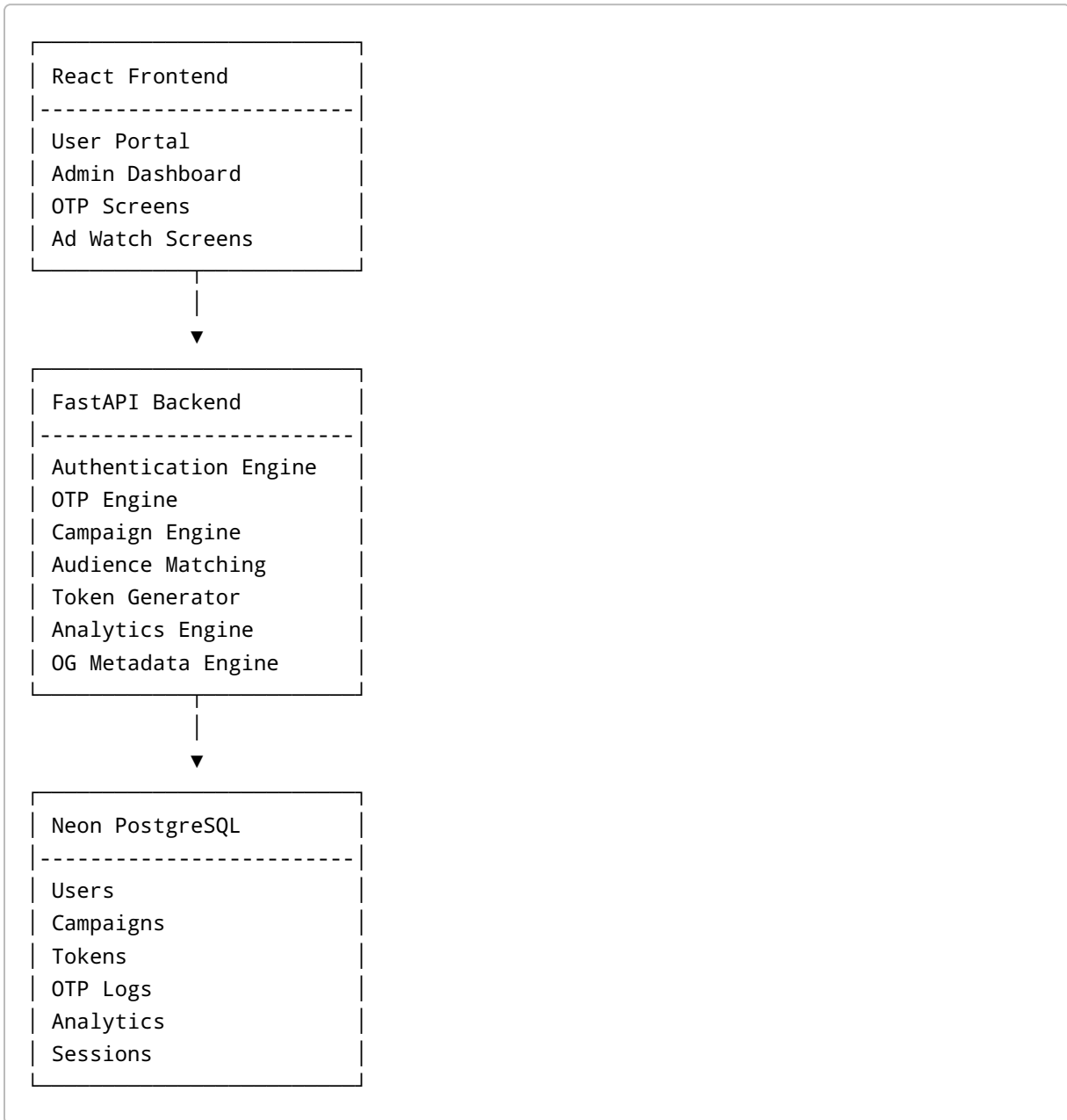
Main Concept

Want OTP?
Watch Advertisement First.

Purpose

- Monetized authentication
- Advertisement engagement
- Multi-surface advertising
- OTP-driven ad experiences

7. SYSTEM ARCHITECTURE



8. TECHNOLOGY STACK

Frontend

Technology	Purpose
React	Frontend Framework
Tailwind CSS	Styling
React Router	Routing
Axios	API Communication
Shadcn UI (Optional)	Premium UI Components

Backend

Technology	Purpose
FastAPI	API Backend
SQLAlchemy	ORM
Pydantic	Validation
JWT	Authentication
Uvicorn	ASGI Server

Database

Technology	Purpose
Neon PostgreSQL	Cloud Database

Deployment

Platform	Usage
Render	Frontend + Backend Hosting

Platform	Usage
Neon	PostgreSQL Hosting
GitHub	Source Code Repository

Optional Technologies

Technology	Purpose
Twilio	OTP SMS Service
Fast2SMS	OTP Service
MSG91	SMS Integration
Rust + WASM	Encryption Demo

9. DATABASE DESIGN

users

Column	Type
id	Integer
name	String
mobile	String
age	Integer
gender	String
area	String

campaigns

Column	Type
id	Integer
name	String

Column	Type
title_template	String
description	Text
image_url	String

targeting_rules

Column	Type
campaign_id	Integer
min_age	Integer
max_age	Integer
gender	String
area	String

generated_tokens

Column	Type
token	String
user_id	Integer
campaign_id	Integer

otp_logs

Column	Type
id	Integer
user_id	Integer
otp	String
status	String

analytics_events

Column	Type
id	Integer
event_type	String
token	String
timestamp	DateTime

sessions

Column	Type
id	Integer
user_id	Integer
jwt_token	String

10. MAIN MODULES

1. User Registration Module

Responsible for:

- User onboarding
- Self registration
- Admin user creation

2. Campaign Management Module

Responsible for:

- Campaign creation
- Creative uploads
- Targeting rules
- Campaign management

3. Audience Matching Engine

Responsible for:

- Selecting matching users
 - Filtering based on rules
 - Personalized audience delivery
-

4. Token Generation Engine

Responsible for:

- Secure token generation
 - Personalized URLs
 - Link validation
-

5. Dynamic OG Metadata Engine

Responsible for:

- Personalized previews
 - Dynamic social media metadata
 - Open Graph rendering
-

6. Ad Delivery Engine

Responsible for:

- Ad rendering
 - Ad rotation
 - Multi-surface ads
 - Ad targeting
-

7. Ad Completion Tracker

Responsible for:

- Tracking ad watch completion
 - Enabling OTP only after ad completion
 - Preventing OTP bypass
-

8. OTP Engine

Responsible for:

- OTP generation
 - OTP sending
 - OTP validation
-

9. Authentication Engine

Responsible for:

- Session creation
 - JWT authentication
 - Auto-login
-

10. Analytics Engine

Responsible for:

- Event tracking
 - Engagement tracking
 - Conversion tracking
 - User activity metrics
-

11. FRONTEND PAGES

Public Pages

```
/
/register
/login
/get-otp
```

Advertisement Pages

```
/ad-watch
/ad-preview/:token
```

OTP Pages

```
/otp-verification
/otp-confirmation
```

User Portal Pages

```
/dashboard
/offers
/recommendations
/profile
```

Admin Pages

```
/admin
/admin/campaigns
```



```
/admin/users  
/admin/analytics
```

12. BACKEND API DESIGN

User APIs

```
POST /register  
POST /login  
POST /verify-otp
```

Campaign APIs

```
POST /campaign/create  
GET /campaigns
```

Token APIs

```
POST /generate-token  
GET /preview/{token}
```

Advertisement APIs

```
GET /ad/watch  
POST /ad/completed
```

OTP APIs

```
POST /send-otp  
POST /verify-otp
```

Analytics APIs

```
POST /track-event  
GET /analytics
```

13. ANALYTICS EVENTS

Track the following:

```
preview_fetch  
ad_impression  
ad_completed  
otp_requested  
otp_generated  
otp_verified  
login_success  
portal_view  
conversion_event
```

14. DEPLOYMENT ARCHITECTURE

Frontend Deployment

```
GitHub → Render Static Site
```

Backend Deployment

GitHub → Render Web Service

Database Deployment

Neon PostgreSQL Cloud Database

15. DEVELOPMENT PHASES

Phase 1 — Project Setup

- React frontend setup
 - FastAPI backend setup
 - Neon database integration
 - Render deployment setup
-

Phase 2 — User & Authentication

- User registration
 - Mobile login
 - OTP system
 - JWT authentication
-

Phase 3 — Campaign Engine

- Campaign creation
 - Audience targeting
 - Token generation
 - Personalized previews
-

Phase 4 — Advertisement-Gated OTP

- Ad watch screens
 - Ad completion tracker
 - OTP unlock system
-

Phase 5 — Portal & Analytics

- Personalized portal
 - Dashboard
 - Analytics tracking
 - Conversion events
-

Phase 6 — Optional Enhancements

- Encryption demo
 - Rust/WASM
 - Advanced analytics
 - Recommendation engine
-

16. POC SIMULATION NOTES

The following systems can be simulated for the POC:

- OTP delivery
- SMS systems
- Advertisement delivery
- Analytics events
- Social media previews
- Encryption workflows
- Authentication sessions

This keeps the system lightweight while still demonstrating the complete APAD concept.

17. FINAL PROJECT SUMMARY

APAD PLATFORM =

User Registration
+
Audience Targeting
+
Personalized Advertisement Delivery
+
Advertisement-Gated OTP Authentication
+
Multi-Surface Ads
+
Automatic Login
+
Personalized Portal Experience
+
Analytics Tracking

18. FINAL ONE-LINE DESCRIPTION

APAD is an advertisement-gated authentication platform where users must complete viewing personalized advertisements before receiving OTP authentication and accessing personalized portal experiences.